

Functional & Performance Testing Template

Model Performance Test

Date	21 FEB 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - POCKET SMART AI
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., Scenario Preferences, Item Names)	Enter valid text strings and special characters/blank spaces in the preference input fields.	Valid inputs are accepted smoothly; meaningful error warnings display for empty or invalid inputs.	Inputs accepted; missing inputs trigger Streamlit warning banners.	Pass
FT-02	Number Input Validation (e.g., Budget ceiling amount, transaction amounts)	Enter values within and outside valid ranges (e.g., negative budget values or 0 budget).	System accepts valid numerical budgets; shows explicit error message for negative or non-numeric inputs.	Rejects negative numeric entries and prompts user with numerical range validation.	Pass
FT-03	Content Generation (e.g., AI Shopping Recommendations & Budget Package)	Provide complete scenario inputs (Budget, Room/Event Type) and click "Generate Recommendation" .	Itemized price breakdown across platforms (Amazon, IKEA, Swiggy) is correctly generated based on input criteria.	Correct budget-compliant recommendation package generated and formatted into tables.	Pass

FT-04	API Connection Check	Enter Google Gemini API key into sidebar and initiate key verification call.	Gemini API authenticates successfully and returns active session confirmation.	google-genai SDK verifies key and initializes model session without errors.	Pass
PT-01	Response Time Test	Measure processing duration from clicking "Generate" to rendering full AI recommendations.	System generates and displays response in under 3 seconds.	Response received and rendered in approximately 1.8 to 2.4 seconds.	Pass
PT-02	API Speed & Concurrency Test	Execute multiple scenario generation queries in rapid succession.	API maintains fast latency without throttling, dropping connections, or slowing down	Subsequent requests processed smoothly with consistent latency under peak load.	Pass
PT-03	Data Batch Processing Load Test	Log large sets of transaction entries into the Core Expense Ledger simultaneously.	Pandas DataFrame updates instantly; Plotly pie charts render smoothly without application crash.	Transactions logged and aggregated instantaneously without UI freezing or crashes.	Pass